Bell Ringer · Solving Equations

ANSWER KEY

Remember: distribute first, then combine like terms. Get the variable terms on one side. Then undo + or − before × or ÷. Whatever you do to one side, do to the other. **Check** by putting your answer back in.

Solve for x. Show your work.

ONE-STEP

1. $x - 7 = 12$

$x - 7 + 7 = 12 + 7$

x = 19

ONE-STEP

2. $-4x = 28$

$-4x \div (-4) = 28 \div (-4)$

x = −7

TWO-STEP

3. $3x + 5 = 26$

$3x = 21 \rightarrow x = 21 \div 3$

x = 7

TWO-STEP

4. $\frac{x}{2} - 6 = 3$

$\frac{x}{2} = 9 \rightarrow x = 9 \cdot 2$

x = 18

DISTRIBUTIVE PROPERTY

5. $4(x - 3) = 20$

$4x - 12 = 20 \rightarrow 4x = 32$

x = 8

DISTRIBUTIVE PROPERTY

6. $-2(x + 5) = 6$

$-2x - 10 = 6 \rightarrow -2x = 16$

x = −8

VARIABLES ON BOTH SIDES

7. $5x - 4 = 2x + 11$

$3x - 4 = 11 \rightarrow 3x = 15$

x = 5

MULTI-STEP

8. $2(3x - 1) + 4 = 4x + 10$

$6x + 2 = 4x + 10 \rightarrow 2x = 8$

x = 4